

The Blueprint for an Organic Revolution

Hempoxies: A Post-Predatory Framework for Advanced Materials



“The transition from a petrochemical-based industrial economy to a regenerative, bio-based ecosystem represents one of the most significant challenges and opportunities of the 21st century.”

The Petrochemical Paradigm is a Dead End

For a century, materials science has been dominated by a “predatory” model: extracting finite resources to create permanent, unrecyclable materials.

This linear “take-make-waste” economy generates persistent pollution and ties our most critical industries to volatile, geopolitically sensitive supply chains.

Thermosets

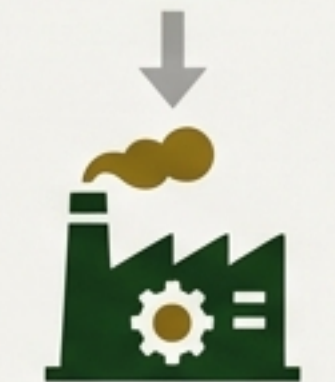
Traditionally, high-performance materials like epoxies are “thermosets”—chemically cross-linked into a permanent structure. They are strong, but once cured, they cannot be melted, reshaped, or recycled. Heating leads to degradation, not melting.



EXTRACTION



SYNTHESIS



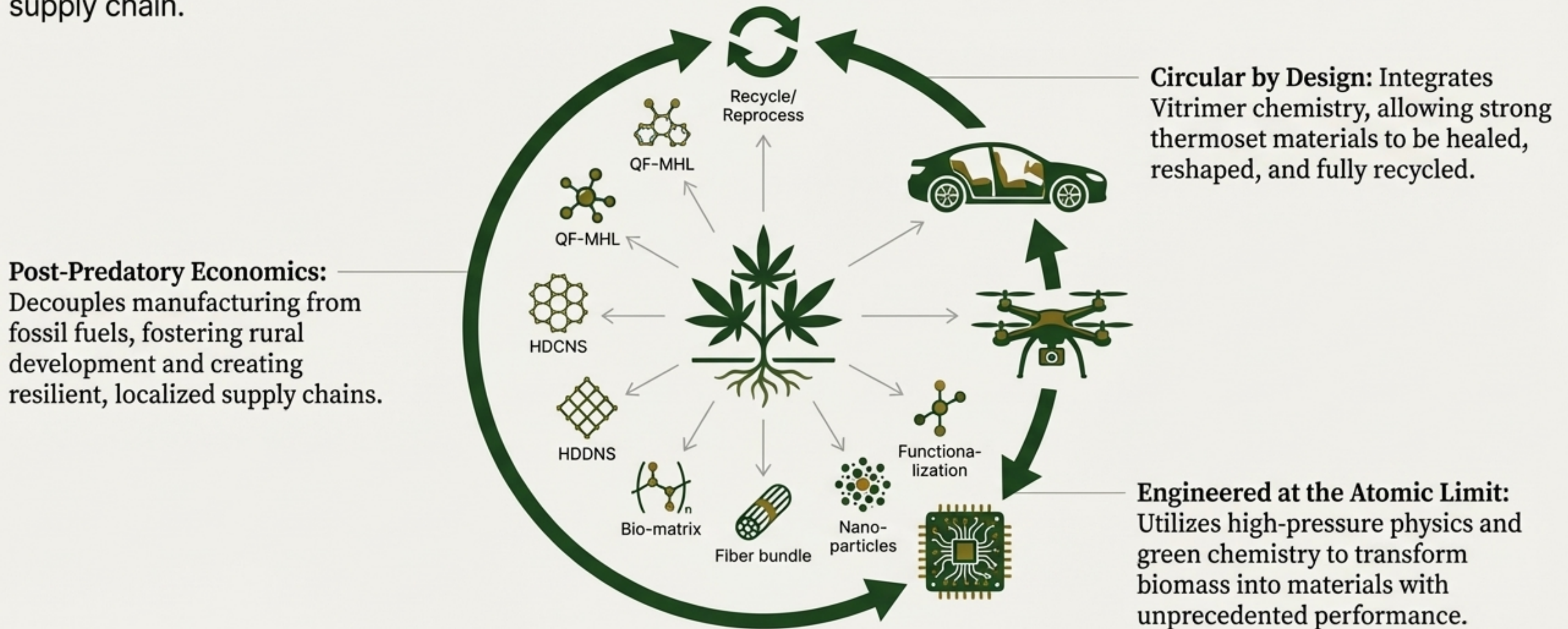
MANUFACTURING



LANDFILL &
OCEAN WASTE

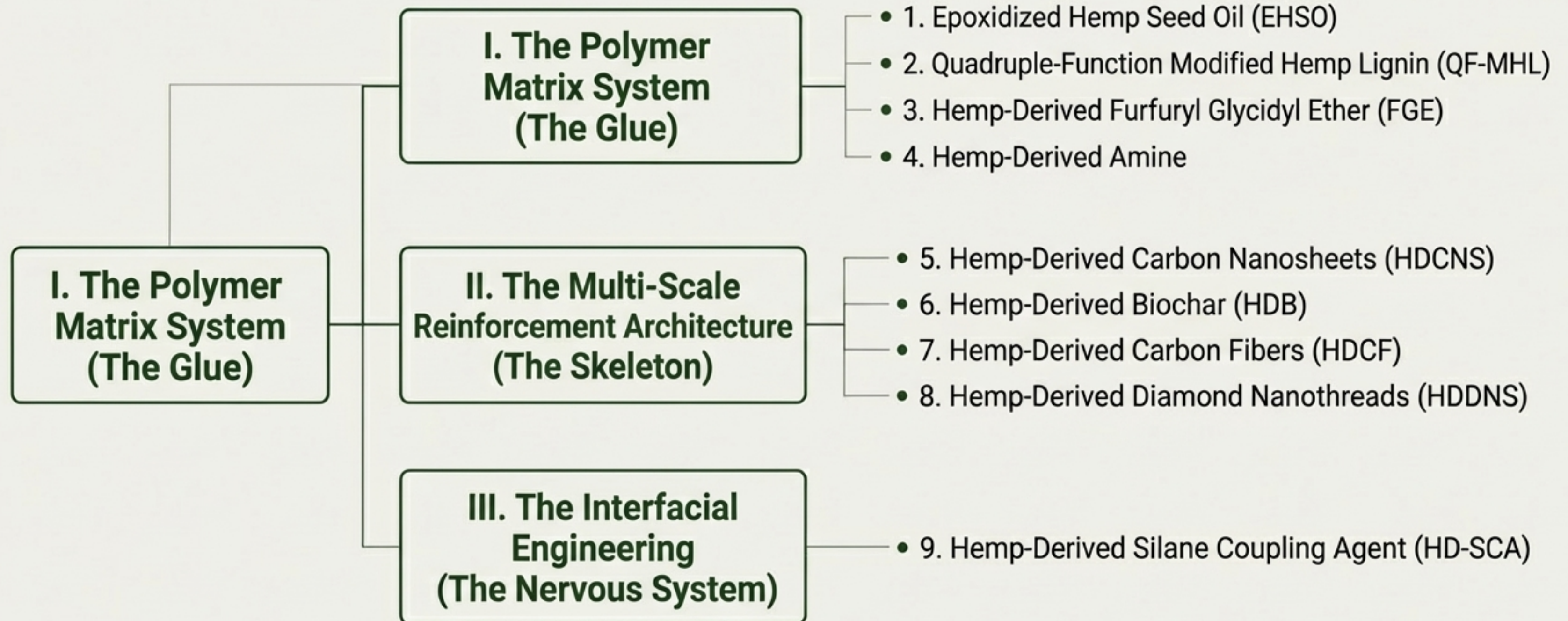
Hempoxies: A Regenerative System, Engineered from a Single Source

Hempoxies is a **9-component bio-nanocomposite** system where every constituent—from the polymer matrix to the ultimate reinforcement—is derived from industrial hemp. This creates a vertically integrated, carbon-negative supply chain.



An Integrated, Multi-Scale Architecture

The performance of Hempoxies arises from meticulous engineering across three integrated subsystems.



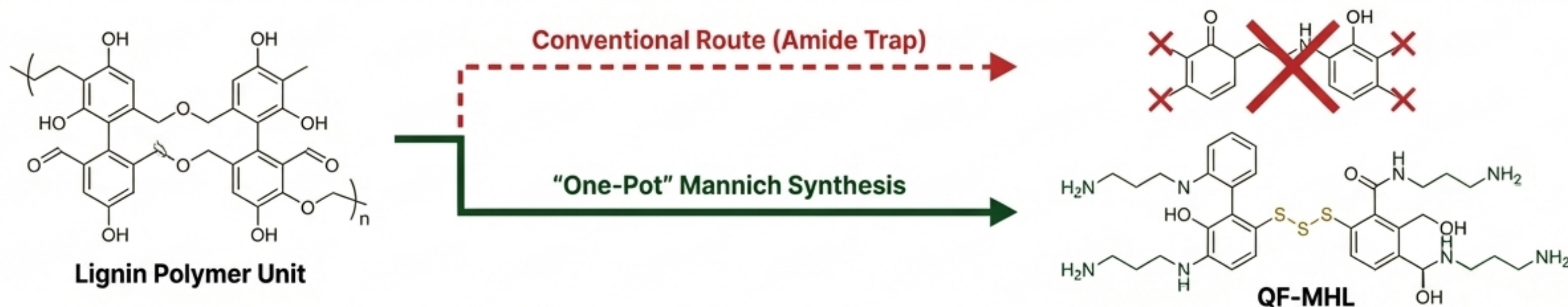
The Chemical Engine: Quadruple-Function Modified Hemp Lignin (QF-MHL)

The Challenge

Lignin, an abundant biopolymer, is notoriously difficult to functionalize. Previous methods like maleinization create an "Amide Trap," where amine hardeners react with the wrong part of the lignin, creating a non-reactive dead end.

The Innovation

We developed a catalyst-free, atom-efficient Mannich reaction that directly grafts cystamine onto lignin's aromatic ring. This elegant pathway avoids the "Amide Trap" entirely and preserves essential molecular structures for quadruple functionality.



1. Hardener: Grafted amine groups cross-link the epoxy matrix.

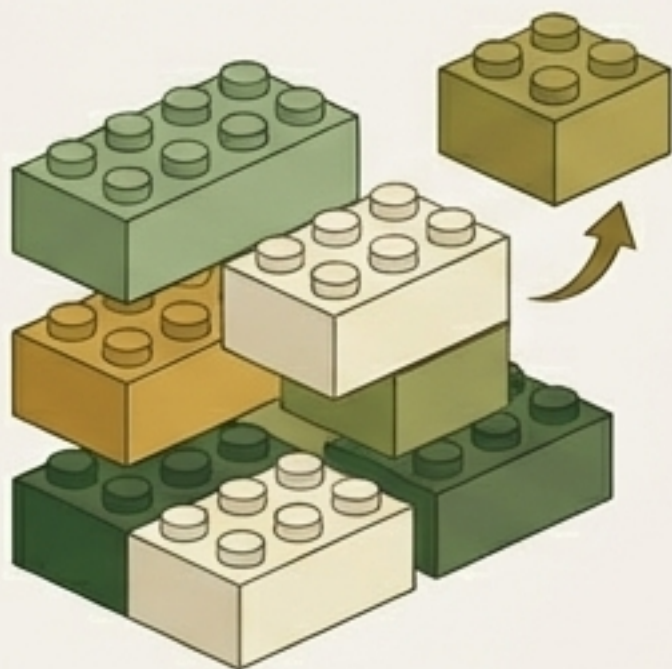
2. Vitrimer Catalyst: Integrated disulfide bonds (S-S) enable dynamic network rearrangement for recyclability.

3. Nano-reinforcement: The rigid lignin core increases material stiffness.

4. Antioxidant: Preserved phenolic hydroxyls act as radical scavengers, preventing degradation.

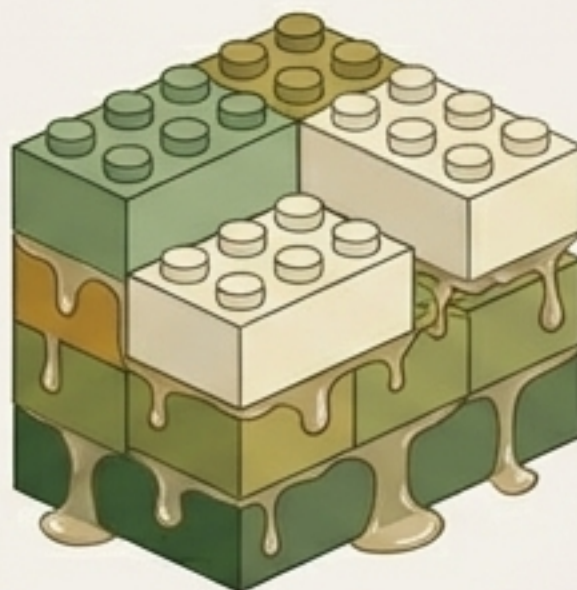
Strong **and** Recyclable: The Power of Vitrimer Chemistry

Thermoplastics are like stacked LEGOs



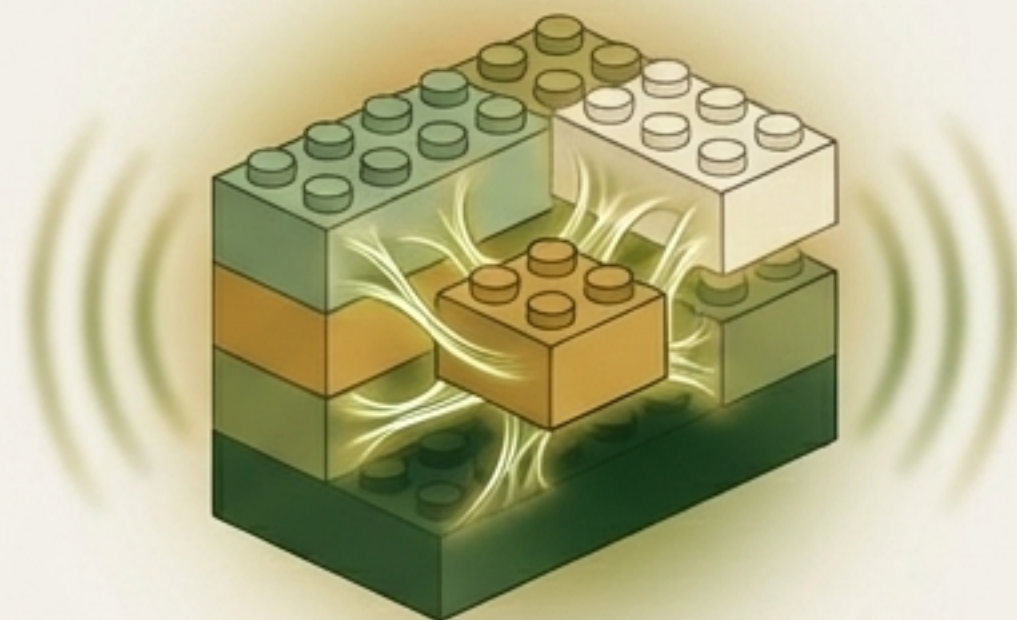
Easy to pull apart and rebuild (recyclable), but not very strong.

Thermosets are like glued LEGOs



Super strong and stable, but you can never take them apart again (not recyclable).

Vitrimers are like LEGOs with "Smart Glue"



The bricks are strongly glued together. But when heated, the glue softens just enough to let the bricks swap places without falling apart. When it cools, the glue hardens again, locking in the new, strong shape.

The Result: Hempoxies combine the strength and solvent resistance of a thermoset with the reprocessability and healability of a thermoplastic.

The Diamond Spine: Hemp-Derived Diamond Nanothreads (HDDNS)

What They Are

The ultimate 1D reinforcement. HDDNS are one-dimensional carbon allotropes with the sp^3 bonding structure of diamond, not the sp^2 structure of carbon nanotubes. This tetrahedral bonding provides unparalleled stiffness and strength.

Performance Metrics

>850 GPa

Young's Modulus
(Stiffness)

~134 GPa

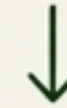
Ultimate Tensile Strength
(Potentially double the strongest carbon nanotubes)



Green Synthesis Route

1. Biorefining

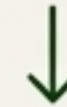
Hemp biomass is converted into pure aromatic precursor crystals via Hydrothermal Liquefaction (HTL) and Catalytic Hydrodeoxygenation (HDO).



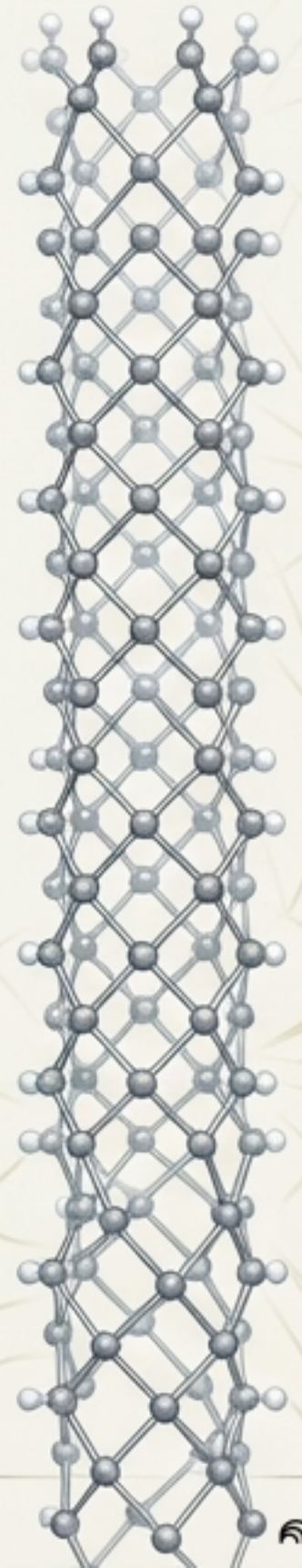
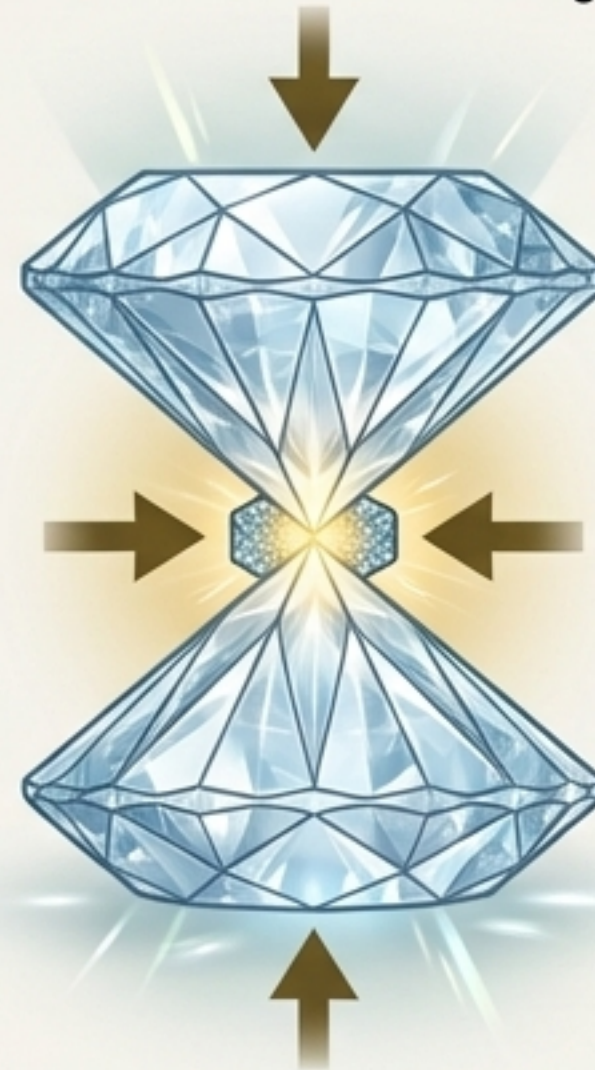
2. Pressure-Induced Polymerization

20 GPa

≈200,000x
atmospheric pressure



Diamond Nanothread



The Linchpin: Engineering the Interface for Maximum Load Transfer

The "Weak Link" Problem

A composite is only as strong as the bond between its matrix and reinforcements. Even with 134 GPa nanothreads, the composite will fail if stress isn't transferred efficiently from the matrix to the threads. Without a chemical bond, fibers simply pull out under load.

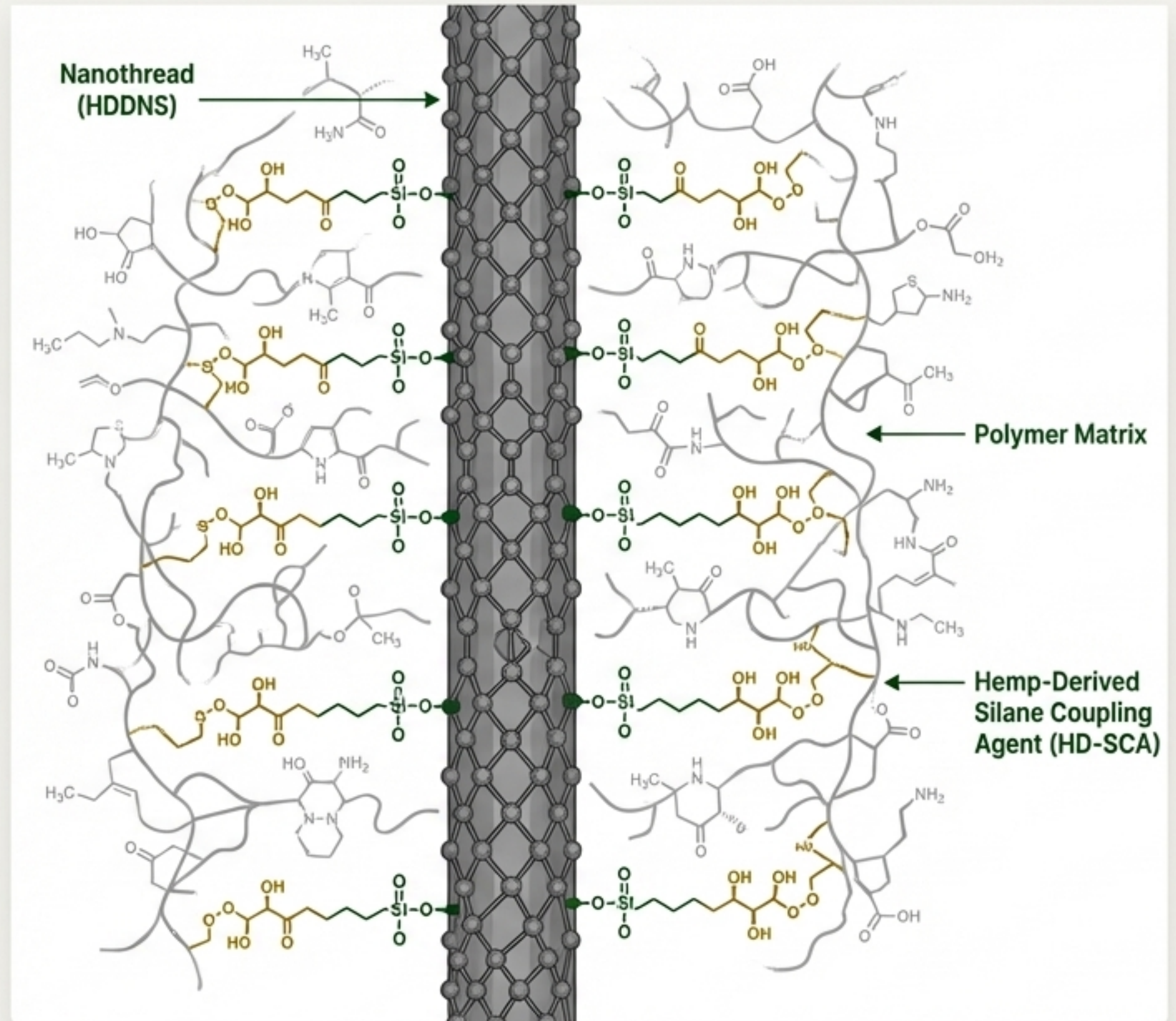
Without a chemical bond, fibers simply pull out under load.

The Solution: A Dedicated Molecular Bridge (Component 9)

The Hemp-Derived Silane Coupling Agent (HD-SCA) is a bifunctional molecule that solves this problem.

- **Inorganic End:** Covalently bonds to the surface of the carbon reinforcements (HDDNS, HDCF, etc.).
- **Organic End:** Co-polymerizes with the epoxy matrix during curing.

The Result: A continuous chain of strong covalent bonds from the matrix, through the silane bridge, to the diamond core of the nanothread. This ensures the full, theoretical strength of the reinforcements is utilized.



Key Performance Metrics & Process Controls

HDDNS Synthesis Pressure

20 GPa

(~200,000x Atmospheric Pressure)

HDDNS Young's Modulus

>850 GPa

(Stiffness)

HDDNS Tensile Strength

~134 GPa

(Strength)

Matrix Target OOC

>6.5%

(Ensures sufficient cross-link density for a rigid matrix)

EHSO Synthesis Control

<60°C

(Critical to prevent premature ring-opening and loss of reactivity)

QF-MHL Synthesis Control

60°C

(Optimal temperature for the 'One-Pot' Mannich reaction)

This multi-scale system—from the molecular engineering of the matrix to the sub-nanometer reinforcement—is designed to arrest crack propagation at every level.

Strategic Landscape: A Clear-Eyed View

Strengths

- Superior theoretical mechanical properties (HDDNS).
- Full recyclability via Vitrimer chemistry.
- Carbon-negative, vertically integrated 'All-Hemp' supply chain.
- Efficient 'One-Pot' synthesis for the QF-MHL matrix.

Weaknesses

- **HDDNS synthesis is currently at TRL 3 (lab scale); industrial scale-up is the primary technical and capital barrier.**
- Delicate process controls required for matrix synthesis.

Opportunities

- Disrupt high-performance sectors (aerospace, automotive).
- Lead the rapidly growing circular economy / bio-composite market (Projected to reach \$60.6B by 2028 @ 14.2% CAGR).
- Drive rural economic development through agricultural sourcing.

Threats

- High capital investment required for high-pressure synthesis scale-up.
- Competition from entrenched petrochemical incumbents.

The Roadmap: From Lab-Scale Breakthrough to Industrial Reality

Present (TRL 3)

- **Focus:** Lab-scale synthesis of all 9 components.
- **Key Achievement:** Proof-of-concept for HDDNS synthesis in Diamond Anvil Cells. Full formulation defined.

Mid-Future: Industrial Commercialization

- **Focus:** Market penetration in automotive, aerospace, and electronics.
- **Key Milestone:** High-volume, continuous HDDNS production. Establishment of robust, vertically integrated agricultural and manufacturing supply chains.

Near Future: Pilot Scale-Up (Investment Focus)

- **Focus:** Empirical validation of cure kinetics and composite performance.
- **Key Milestone:** Scale precursor biorefining to continuous flow reactors. Develop initial macroscopic HDDNS production using modified Belt-Type or Toroidal Presses.

Long-Term Vision: The Organic Revolution

- **Focus:** Hempoxies as a cornerstone of global sustainable infrastructure.
- **Key Milestone:** A fully circular, carbon-negative industrial ecosystem.

Action Plan: The Next Steps for the Hempoxies Revolution

We are seeking strategic partners and investment to execute a focused plan to advance Hempoxies from TRL 3 to commercial readiness.

Immediate Priorities

1. **Industrial Scale-Up of HDDNS**

Invest in R&D and capital equipment (e.g., Toroidal Presses) to transition high-pressure synthesis from microscopic to macroscopic scale. **This is the #1 priority.**

2. **Empirical Validation & Optimization**

Fund rigorous testing of the full 9-component composite to validate mechanical, thermal, and long-term performance against aerospace standards.

3. **Continuous Biorefining Development**

Establish pilot programs for continuous flow reactors to ensure a scalable and cost-effective supply chain for all hemp-derived precursors.

4. **Strategic Partnerships**

Forge alliances with key players in aerospace and automotive to co-develop application-specific formulations and secure early adopters.

The Visionary and the Ecosystem

Creator



Marie-Soleil Seshat Landry

Creator of Hempoxies, visionary for the “Post-Predatory” economic model, and CEO of Landry Industries.

Key Voices & Research in the Field

Prof. David Mitlin (University of Alberta)

Pioneering work on hemp-derived carbon nanosheets.

Senay Simsek, PhD (Purdue University)

Research on sustainable, biodegradable superabsorbent materials from hemp.

Ronald Kander, PhD (Thomas Jefferson University)

Expertise in bio-composites and sustainable materials.

Frequently Asked Questions

Q What makes Hempoxies different from other bio-composites?

A Two things: 1) The 'All-Hemp' vertically integrated system, and 2) Unprecedented performance from Hemp-Derived Diamond Nanothreads (HDDNS), which moves it from a 'green alternative' to a superior performance material.

Q What is the primary barrier to commercialization?

A The industrial scale-up of HDDNS synthesis. The science is sound at lab scale (TRL 3); the challenge is engineering high-pressure systems for high-volume manufacturing. This is the focus of our next phase.

Q How is it truly circular?

A The Vitrimer chemistry in the matrix. It allows the cross-linked thermoset to be re-molded, healed, or recycled at the end of its life, retaining the material's high value and preventing waste.

Q Is this carbon-negative?

A The model is designed to be carbon-negative. Industrial hemp is a highly efficient carbon sequestration crop. By locking that carbon into durable, long-life products and displacing petrochemicals, the full lifecycle can sequester more CO₂ than it emits. A formal Life Cycle Assessment (LCA) is a key part of our action plan.

Engineering at the Atomic Limit for a Regenerative Future

“The era of the ‘Organic Revolution’ is not just about growing materials; it is about engineering them at the atomic limit.”

— Marie-Soleil Seshat Landry

Hempoxies are the blueprint for that future.